

Evaluating Representations in Dimension Reduction Using CLID Metric

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Abstract. This project investigates label-free evaluation of dimensionality reduction techniques using the CLID metric, which combines Cluster Learnability (CL) and Intrinsic Dimension (ID) to assess the quality of representations without relying on supervised tasks. The analysis focuses on three methods: PCA, t-SNE, and UMAP, applied to two real-world tabular datasets: Adult and Bank Marketing. CLID scores were computed for 2D and 3D outputs across all methods, while 5D and 10D was only evaluated for PCA and UMAP due to the high computational cost of t-SNE in higher dimensions. The results show that PCA consistently achieved the highest average CLID scores, indicating strong structure retention and low intrinsic dimensionality. UMAP follows, and is particularly efficient on the Bank Marketing dataset, although with slightly higher intrinsic dimensionality. t-SNE showed the lowest CLID scores and highest variability, especially in 3D, suggesting less stable output representations.

1 Introduction

Evaluating the quality of learned representations, especially when labeled data is limited or unavailable is a big challenges in machine learning. Many standard evaluation procedures first train a classifier on top of fixed features and use accuracy as the main score, but that process often adds extra modeling decisions and hyperparameter choices that can mask what the representation itself is contributing. This approach introduces limitations. Model choice, training stability, hyperparameter tuning, and data set splits can all affect the results. As a consequence, it is difficult to conclude whether good performance comes from the representation itself or from the model setup.

These issues grow in self-supervised learning (SSL), where the representations are learned without labels and must be evaluated post-hoc. Downstream tasks reduce the possibility to compare representations directly.

Evaluation methods such as "Uniformity and Alignment" and "RankMe" have been discussed in literature. Wang and Isola [1] introduced the concepts of alignment and uniformity as measurable properties that explain why contrastive representation learning produces effective features. Building on the idea of label-free evaluation, Garrido et al. [2] introduced RankMe, a metric that assesses the quality of joint-embedding self-supervised representations across different downstream datasets without requiring labeled data.

Internal evaluation metrics help by assessing the structure and usefulness of representations without requiring labels. One such metric, CLID [3] combines two aspects of representation quality: Cluster Learnability (CL), and Intrinsic Dimension (ID) capturing both how well a representation supports simple cluster prediction and how compact its structure is.

Although internal metrics such as CLID have been proposed for general representation learning, they have rarely been applied to the outputs of dimension reduction (DR) methods, as DR evaluation typically relies on projection-distortion criteria such as trustworthiness, continuity,

and co-ranking based measures [4]. These methods, such as PCA [5], t-SNE [6], and UMAP [7] are commonly used to obtain lower-dimensional representations for visualization or analysis. They compress high-dimensional data into a smaller number of features, often used directly for interpretation, clustering, or further processing. However, these outputs are usually estimated qualitatively (e.g., via 2D plots). It would be useful to measure their internal quality in a structured, label-free way.

This project will be focused on bridging this literature gap by using CLID to evaluate the internal quality of the reduced representations obtained from three commonly used DR techniques—PCA, t-SNE, and UMAP. CLID will be used across multiple output dimensions, including 2D, 3D for all three methods, and 5D and 10D for PCA and UMAP due to the high computational cost of t-SNE in higher dimensions. This approach will be applied on two UCI tabular datasets: Adult [8] and Bank Marketing [9]. The idea is to compute CLID for low-dimensional embeddings from each DR method and compare how well these methods preserve structure in their reduced representations. By analyzing how CLID varies with dimensionality, the project will explore how increasing or decreasing the number of retained dimensions affects the internal structure and learnability of the resulting representations.

2 Data and Methods

2.1 Datasets and Preprocessing

In the project two tabular datasets from the UCI repository are used: Adult [8] and Bank Marketing [9]. Both datasets include a mix of numerical and categorical variables, and the data needed some preprocessing steps before dimension reduction is applied.

The preprocessing has been completed through a shared pipeline between datasets and converts each dataset into a fully numeric feature matrix X . A fixed global seed (RANDOM SEED = 42) is used for reproducibility.

As the Adult dataset contained missing values (in the dataset description labeled with "?"), the rows containing such values were removed. The categorical features were encoded with one-hot encoding, preparing them for the model fitting phase. The numerical features were normalized with standard scaling so that distance-based methods such t-SNE, UMAP, KNN (K-nearest neighbors) behave consistently across feature types. Lastly, the output is a single numeric feature matrix X for each dataset, which was then used as the input to all DR methods.

2.2 CLID Metric

For a given reduced representation $Z \in \mathbb{R}^{n \times d}$, the evaluation computes two components:

1) Cluster Learnability (CL)

CL is assessed by first creating pseudo-labels $\tilde{y}$ by clustering the representation Z with K-means algorithm. We then split $(Z, \tilde{y})$ into a training set and a held-out validation set (70/30) and train a k -nearest neighbors classifier on the training split to predict $\tilde{y}$. Finally, CL is defined as the classification accuracy on the held-out split - the fraction of validation points whose predicted pseudo-label matches the K-means label. In our experiments, we use $K = 10$ clusters and $k = 5$ neighbors.

2) Intrinsic Dimension (ID)

The project estimates intrinsic dimension using the TwoNN estimator, which relies on the ratio between the distances to the first and second nearest neighbors to quantify the effective dimen-

sionality of the representation manifold. The implementation uses the maximum likelihood (MLE) form of TwoNN and computes ID as $\hat{d} = \left(\frac{1}{N} \sum_{i=1}^N \log \left(\frac{r_{i2}}{r_{i1}} \right) \right)^{-1}$, where r_{i1} and r_{i2} denote the distances from point i to its first and second nearest neighbors. CL and ID can then be combined into a unique metric "CLID" by first standardizing CL and ID and then summing them after scaling: $\text{CLID} = \text{CL} + \text{ID}$.

This combination could possibly be a useful metric for comparing how well different DR methods preserve meaningful structure in data.

2.3 Dimension Reduction (DR)

Dimension reduction methods [10] refer to the process of converting high-dimensional data (with many features) into lower-dimensional form (fewer features) while preserving important patterns, relationships, or structures in the data. It helps simplify data, speed up processing, and make visualizations possible. In this project three dimension reduction methods are considered: PCA, t-SNE, and UMAP.

Principal Component Analysis (PCA) [5] finds new axes that captures most of the variance in the data and projects the data onto these axes, producing a linear, compressed version of the original dataset.

t-distributed Stochastic Neighbor Embedding (t-SNE) [6] focuses on preserving local similarities, so that nearby points in the original space stay close in the reduced space; it is nonlinear and mainly used for visualization.

Uniform Manifold Approximation and Projection (UMAP) [7] also preserves local structure but is generally faster than t-SNE and tends to keep more of the global structure, making it suitable for both visualization and downstream tasks.

The idea of the project was to produce representations in multiple output dimensions for each dataset and method. Each output is treated as a representation $Z \in \mathbb{R}^{n \times d}$, where $d \in \{2, 3, 5, 10\}$, and then evaluates Z with **CL**, **ID**, and **CLID**.

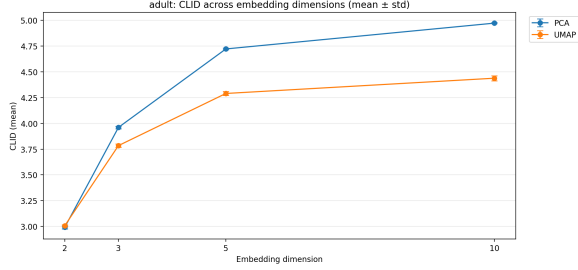
Due to high computational cost of t-SNE in dimensions larger than 3 (in sklearn.manifold.TSNE, method = 'exact' must be used, not approximate optimization via Barnes-Hut), the dimensionality comparisons were constrained to 2 and 3 dimensions for this method, while results for PCA and UMAP are presented in 2D, 3D, 5D and 10D output dimensions. These dimensionalities allow us to evaluate trends in CLID across methods and datasets, and are relevant for real-world applications where embeddings are used for visualization or low-dimensional analysis.

2.4 Experimental Setup

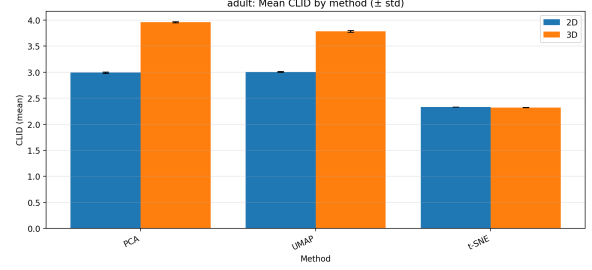
For each dataset (Adult and Bank Marketing), method (PCA, t-SNE, and UMAP) and dimension combination, 10 trials have been done and CL, ID, and CLID were recorded for each run. PCA and UMAP are evaluated in 2D, 3D, 5D, and 10D, while t-SNE is restricted to 2D and 3D due to the Barnes-Hut implementation constraint. Each trial uses a different random seed (`RANDOM_SEED + run`) to capture run-to-run variability for stochastic methods. For t-SNE and PCA, the implementation adds small Gaussian noise to the final representations, and for PCA, it also adds slight random noise to the input data before applying the method. This introduces controlled variability, ensuring that the results vary slightly across runs. The results are summarized by reporting the mean and standard deviation of CL, ID, and CLID across the 10 runs for each setup.

3 Discussion

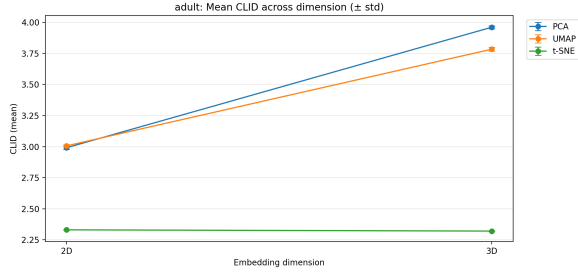
The experiments compare how PCA, t-SNE, and UMAP present reduced representations on the Adult and Bank Marketing datasets. The evaluation uses CLID in 2D and 3D, and it also tracks how CLID changes as the embedding dimension increases (5D and 10D) for PCA and UMAP. Figures 1a–1d summarize results on Adult, and Figures 2a–2d summarize results on Bank Marketing dataset. The plots report mean performance and variance across repeated runs.



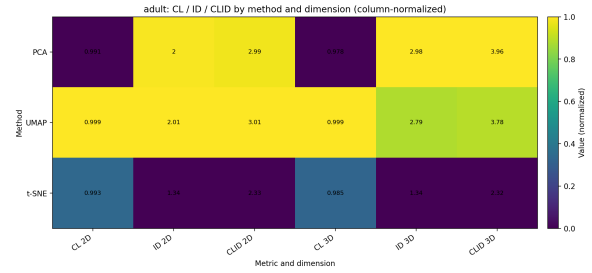
(a) Adult: CLID across embedding dimensions (mean $\pm$ std) for PCA and UMAP.



(b) Adult: mean CLID by method in 2D and 3D (mean $\pm$ std).



(c) Adult: mean CLID across 2D and 3D by method (mean $\pm$ std).



(d) Adult: CL, ID, and CLID by method and dimension (column-normalized).

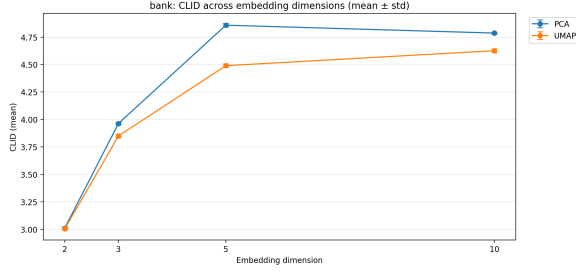
Figure 1: Adult dataset visualizations.

Across all runs, PCA achieved the highest overall CLID scores when averaged across datasets and dimensions (Figures 1b and 2b). This result suggests that despite being a linear method, PCA retains a structure that supports strong KNN classification performance on K-means labels and exhibits low ID. UMAP showed strong clustering ability but slightly higher estimated ID, which reduced its total CLID (Figures 1b and 2b). t-SNE, showed lower overall CLID scores (Figures 1b and 2b) due to reduced CL, especially in 3D.

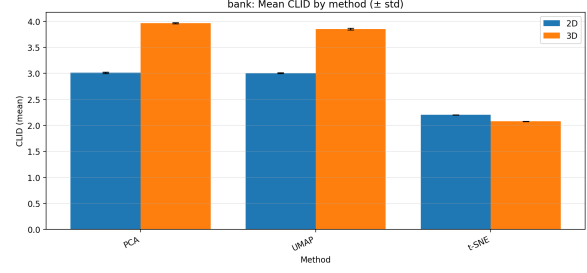
t-SNE relies on randomness and focuses mainly on local relationships, therefore, it does not always produce representations that are consistent or well-structured.

Comparing across the datasets shows smaller differences. On Adult, UMAP barely exceeds PCA in 2D, while PCA appeared as best in 3D. (Figure 1c). On Bank Marketing, PCA is slightly ahead of UMAP in both 2D and 3D.(Figure 2c).

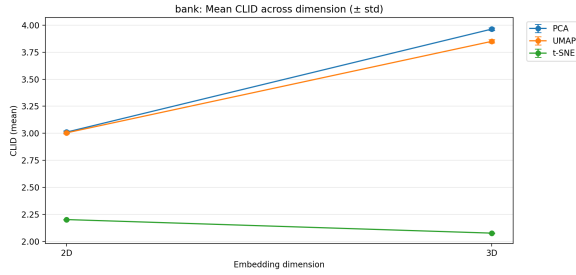
Figures 1d and 2d show the mean values of CL, ID, and CLID for each method in 2D and 3D. The heatmaps use column normalization, so the colors compare methods within the same metric and dimension. In both datasets, CL stays very high for all methods, meaning that KNN can predict the K-means pseudo-labels well from each embedding. The biggest differences come



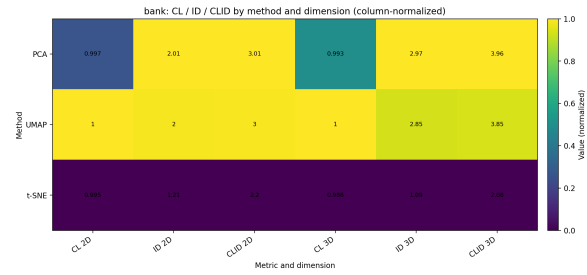
(a) Bank Marketing: CLID across embedding dimensions (mean $\pm$ std) for PCA and UMAP.



(b) Bank Marketing: mean CLID by method in 2D and 3D (mean $\pm$ std).



(c) Bank Marketing: mean CLID across 2D and 3D by method (mean $\pm$ std).



(d) Bank Marketing: CL, ID, and CLID by method and dimension (column-normalized).

Figure 2: Bank Marketing dataset visualizations.

from the ID: PCA and UMAP have much higher ID than t-SNE in both 2D and 3D. Since CLID is computed as $CL + ID$, the difference in ID values explains why PCA and UMAP achieve higher CLID than t-SNE.

Increasing the dimension has shown an increase in mean CLID for PCA and UMAP. Figures 1a and 2a show large growth in mean CLID from 2D to 5D, followed by smaller changes in 10D. In Bank Marketing dataset, PCA peaks around 5D and then decreases slightly at 10D, while UMAP continues with a slight upward trend. In Adult dataset, both PCA and UMAP continuously grow up to 10D, with highest values for PCA.

Overall, PCA gives the highest and most stable CLID outcomes across datasets and dimensions, UMAP provides a strong nonlinear alternative with similar behavior, and t-SNE shows lower CLID.

4 Conclusion

This project evaluates the quality of representations produced by DR techniques using CLID metric. By applying PCA, t-SNE, and UMAP to two datasets (Adult and Bank Marketing) and running multiple trials across 2D, 3D, 5D, and 10D (where possible), the experiments revealed several consistent trends.

Across both datasets, PCA achieved the highest CLID scores overall. It consistently produced low ID and supported strong KNN recovery of cluster labels, indicating well-separated, low-dimensional structure in its outputs. UMAP shows similar results, with stronger effects in the Bank Marketing dataset, particularly in 2D and 3D, but its slightly higher ID limited its CLID

score relative to PCA. t-SNE performed noticeably worse in all configurations, with low CL and limited number of output dimensions. These results suggest that t-SNE may not be suitable for applications where representation quality needs to be stable and globally interpretable.

Dimensionality also played an important role. For PCA and UMAP, CLID generally increased with dimensionality up to 5D or 10D. In contrast, t-SNE showed little change across 2D and 3D, reflecting its large focus on local neighborhoods. Interestingly, for both datasets, the CLID scores in 2D were already competitive, which supports the use of low-dimensional embeddings for exploratory analysis when label-free evaluation is required.

Overall, the CLID metric provided stable and interpretable feedback on the structural quality of reduced representations, without relying on downstream tasks. The results support its usefulness for internal evaluation of dimensionality reduction pipelines, especially in unsupervised or semi-supervised learning settings.

5 Recommendations for Future Work

Future work could expand this analysis in several directions. First, incorporating additional evaluation metrics such as RankMe or alignment/uniformity scores would provide a richer understanding of representation structure. Second could be extending the study to a broader range of datasets, especially those with higher intrinsic dimensionality or more complex feature interactions. Another possible extension is evaluating deep self-supervised embeddings (e.g., from SimCLR or DINO) before and after dimensionality reduction. This would connect the current work more directly to the field of self-supervised learning and help clarify how much structure is retained when compressing high-dimensional learned features.

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